Machine Learning and Food Security

Dissertation Proposal

Yujun Zhou

Department of Agricultural and Consumer Economics
University of Illinois at Urbana-Champaign

12/16/2019

Dissertation overview

Hunger is on the rise throughout Africa, with famine threatening millions across several countries. Food security is essential issue for social, economics, and political stability. Factors that are central to food security, including the market price of grains, weather shocks in grain production areas, and agricultural policies.

Dissertation Chapters

- Chapter 1. Effects of Stock-holding Policy on Maize Prices: Evidence from Zambia
- Chapter 2. Predicting Food Security with Machine Learning
- Chapter 3. Performance evaluation for forecasting modeling with spatiotemporal structures in data

Chapter 1
Effects of Stock-holding Policy on Maize Prices: Evidence from Zambia

Research Question

- Are stockholding policies effective in stabilizing grain prices over space and time ?
  - Developing countries often intervene to stabilize domestic staple prices (Minot 2011)
  - Many have adopted stockholding policies (Jayne, 2012)
  - e.g. the Food Reserve Agency (FRA) in Zambia, dominant player on maize market

- Can FRA purchases raise prices for surplus maize producers during time of harvest?
  - raise local prices for surplus maize producers about 5% on average at the time of harvest

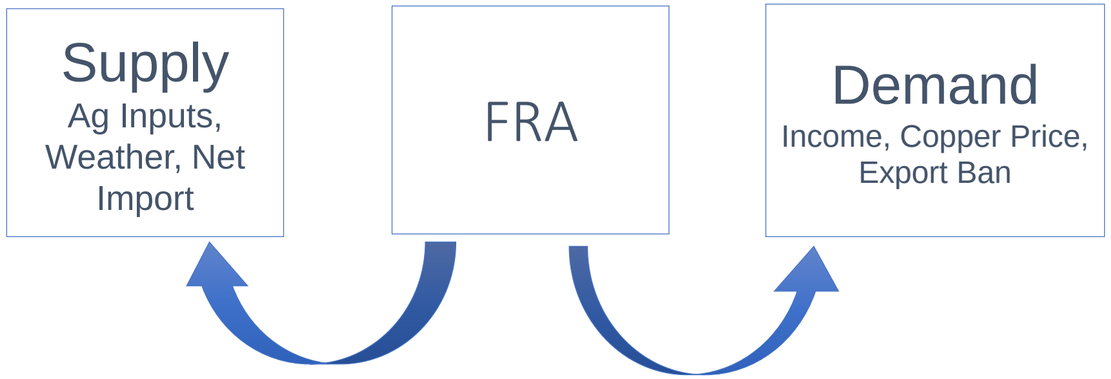
- Can FRA sales mitigate retail price swings associated with domestic production shocks?
  - lower the prices to consumers during the lean season up to 7%

Contribution

- Most studies use VAR-type analysis; Our panel approach over both space and time
- Use instrumental variables to solve the endogeneity in purchase and sales
- Provides empirical evidence on stabilization effect over space and time

Framework

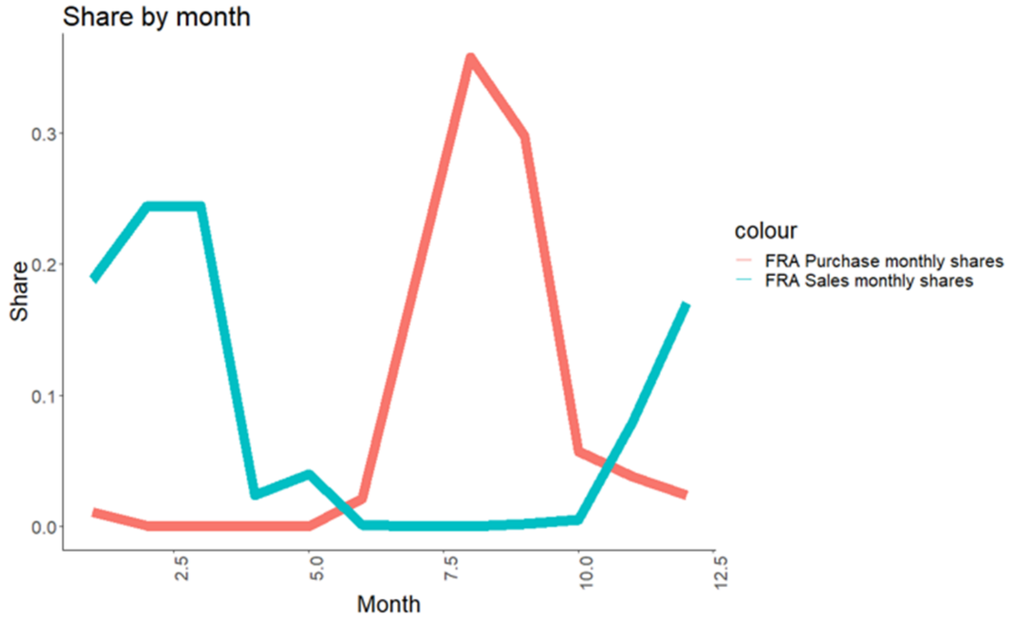
System of Demand and Supply

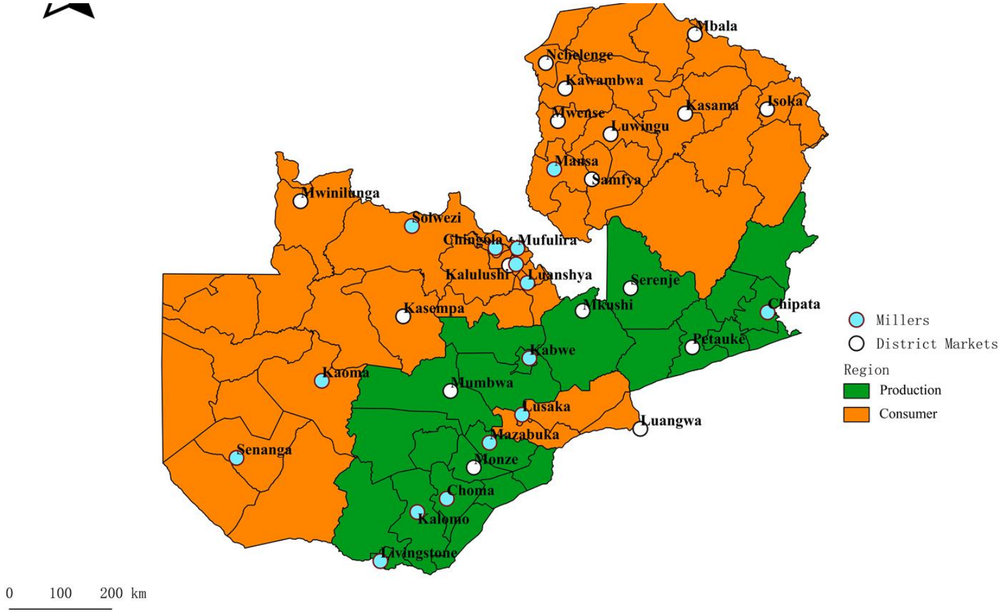


FRA Purchase
(July – October)

FRA Sales
(Dec-March)

Smooth price intra-year, inter-year, and over space





Identification strategy

Purchase
FRA targets surplus areas, more in a bad year

Instrument

Predicted FRA purchases for actual FRA purchase

Underestimate:
places of surplus maize have low price, causing FRA purchases to be spuriously correlated with low prices.

Sales
reverse causality: sell more maize when the price is higher

Instrument

Predicted FRA Sales

Underestimate:
FRA sales positively correlated with prices

Simultaneous policies
Fertilizer Input Subsidy Program, Export Ban, government imports

Control

- year-over-year variations
- lagged price and agriculture-related weather shocks

Overestimate:
Attribute the smoothing effect of these policies on FRA activities

Data

- Monthly Zambia retail maize prices of 32 markets from Jan. 2003 to Dec. 2008
- Annual Zambia FRA purchases from 2002 to 2009 by the district from the FRA, and monthly national total purchase
- Monthly national FRA sales from the FRA
- Agriculturally-relevant precipitation from the Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) during the October–April growing season.
- Temperature data from the African Drought Monitor

Reduced-form model

Following Chapoto and Jayne (2009)

$$Y_{it} = Y_{i,t-1} + FRA_buy_{it}\alpha + FRA_sale_{it}\beta + W_{i,t-1}\gamma + X_{it} + month_t + mkt_i + \varepsilon_{it}$$

Y_{it} is price and price deviations at district i at time t

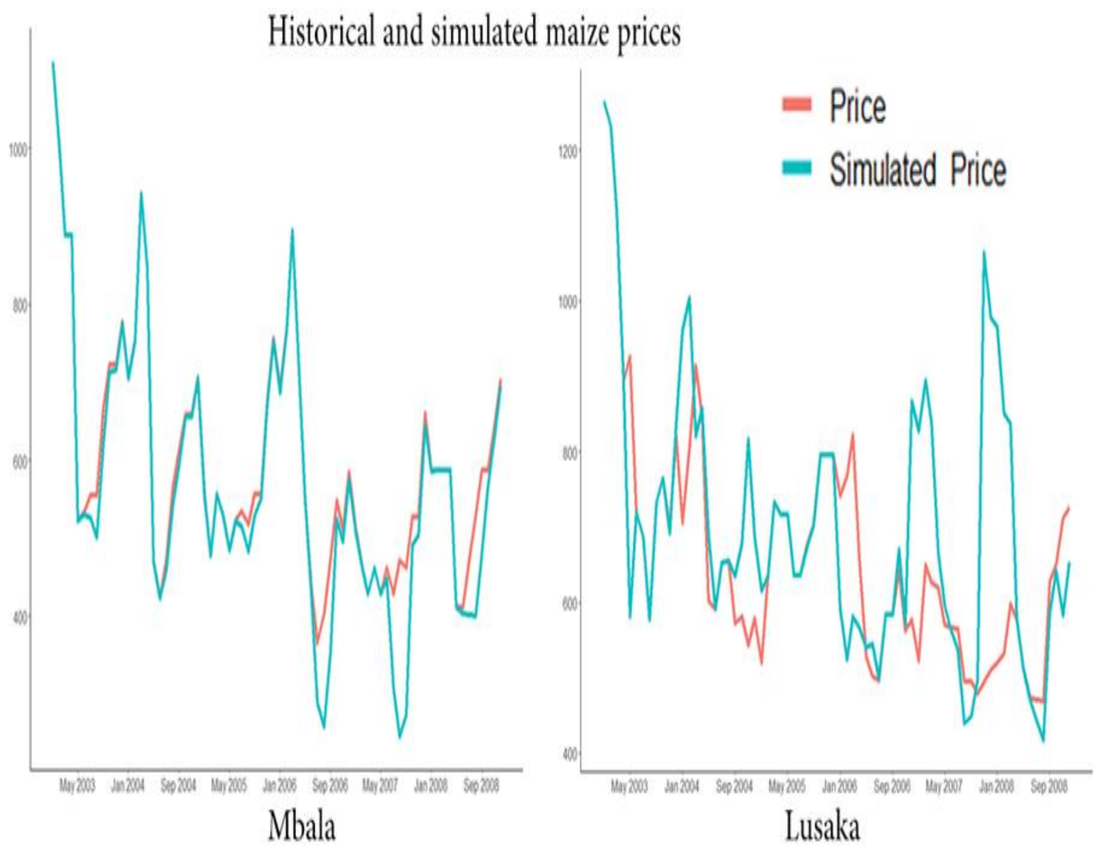
$W_{i,t-1}$ is a vector of weather variables from the previous growing season

X_{it} is a vector of other demand/supply shifter , and ε_{it} is a random error term.

Results

Response Variable:	Maize Price		
	(1)	(2)	(3)
	FE	FE+IV	FE+IV
FRA Purchase	0.020***	0.031***	0.020***
	(0.002)	(0.009)	(0.006)
FRA Sales	-0.458***	-7.908***	-4.531***
	(0.092)	(1.491)	(1.004)
Price Lag			0.554***
			(0.026)
Weather vars	Yes	Yes	No
SAFEX price	Yes	Yes	Yes
Year Fixed Effect	No	No	Yes
Month Fixed Effect	Yes	Yes	Yes
Market Fixed Effect	Yes	Yes	Yes
N	2304	2304	2232
Cluster	32	32	32
Anderson-Canon Corr. LM Statistic	-	33.262	31.851
Cragg-Donald Wald F Statistic	-	16.501	15.823

Simulated price without FRA



Conclusion

- FRA lower price and volatility during the lean seasons and helps support the purchase price that the farmers receive
- smooth both seasonal and spatial variations between the consumption and production regions
- Failing to control for the reverse causality lead to biased estimates
- Consumers in the consumption regions benefit most from the policy
- FRA's effect on between year price variations seems to be rather small.

Chapter 2.
Predicting Food Security with Machine Learning

Research Question

- Can we build an early warning system of food security in areas where data are scarce and data collection is costly? (Hutchinson,1991)
- Successfully detect the rare events food insecurity, informing the when and where food shortages tend to happen
- A framework that is automatically updated, generalizable, scalable and cost-effective

Problem and Opportunity

The problem

- We lack the ability to identify food insecure populations in time to intervene. Humanitarian response tends to trail the onset of food security crises.
- Currently use the Integrated Food Security Phase Classification System (IPC)
- The IPC has large data requirements and has been accused of political influence
- Need to improve prediction of food security crises

The opportunity

- Recent increase in available data related to food security, rainfall, and prices.
- These data are often evaluated in isolation.
- Incorporate these data into a single predictive model of food security early warning.

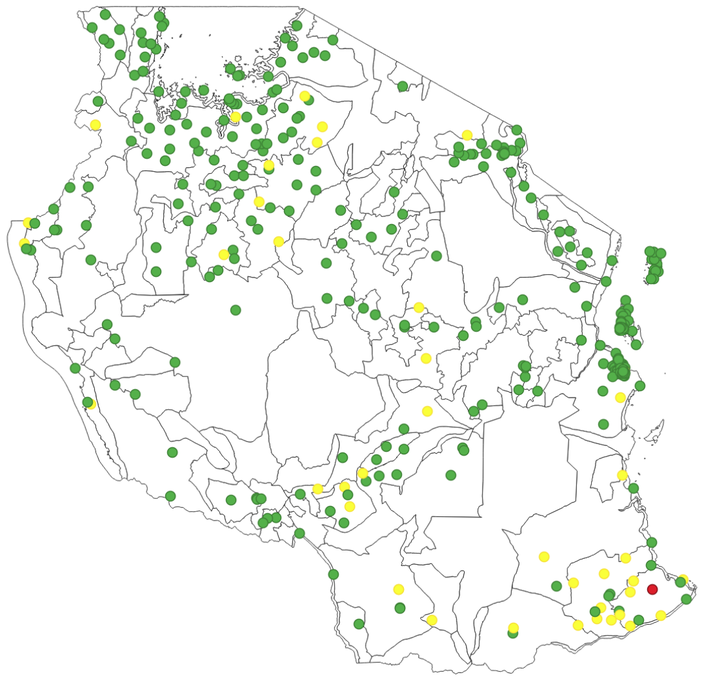
Summary

- Build ML models to predict cluster-level food security status for targeting, aid purposes in times of food shortage
- Use LSMS data for Malawi (2010/11, 2013, 2016) Tanzania (2010/11, 2012/13, 2014/15) and Uganda (2010, 2011, 2012) as ground truth
- Use market price of food staples, weather shocks in growing seasons, and geospatial features around clusters to predict potential food security challenges
- Use data techniques (oversampling, data segmentation) to improve prediction performance
- Correctly categorize 63-84 % of food insecurity categories and up to 20-37% of most food insecure category.

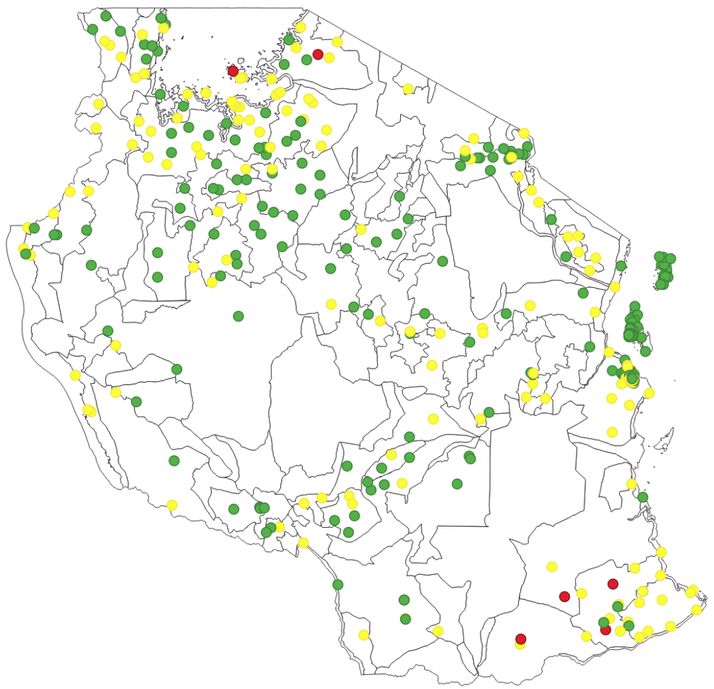
Decisions

1. Categorical versus continuous prediction
2. If categorical, how do we address rare events?
3. What algorithm do we use? And how do we assess it?
4. How do we split the data?

Data

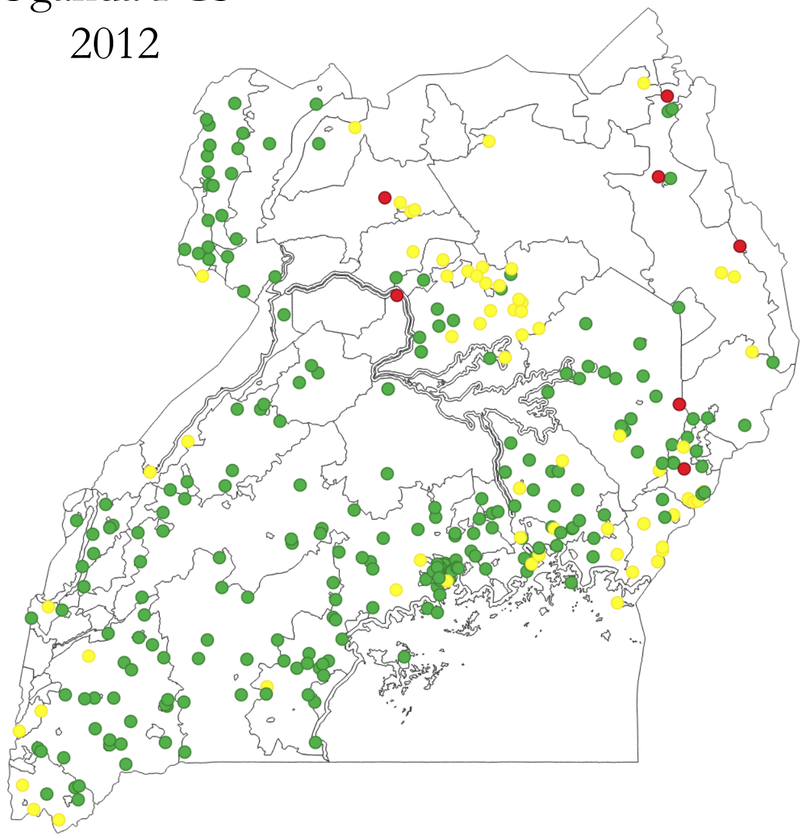


Tanzania FCS
2014/15



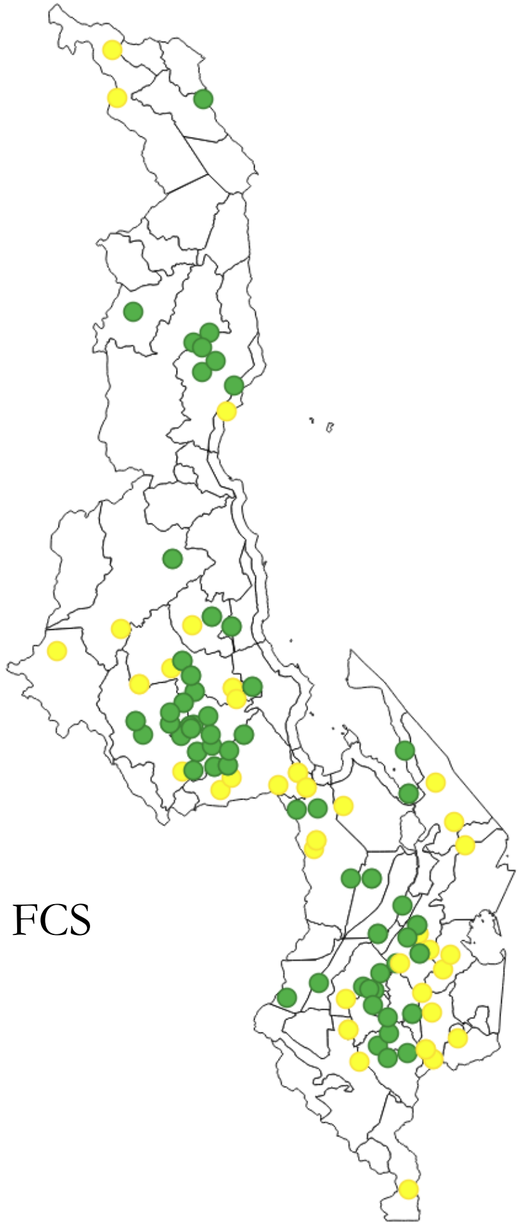
Tanzania rCSI
2014/15

Uganda FCS
2012

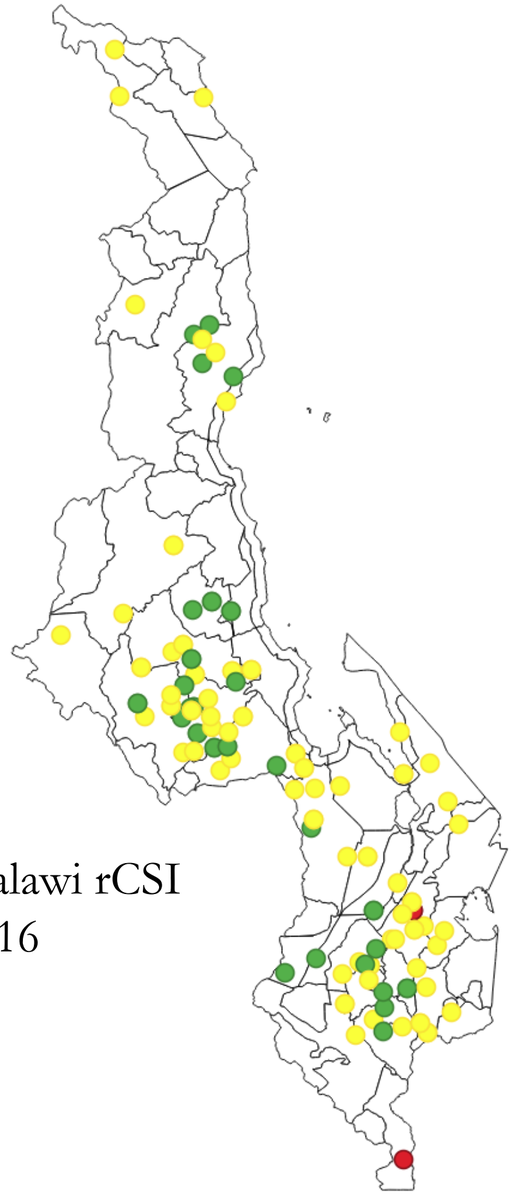


Legend
- High Food Security
- Medium Food Security
- Low Food Security
- Livelihood zones

Data



Malawi FCS
2016



Malawi rCSI
2016

Legend
- High Food Security
- Medium Food Security
- Low Food Security
- Livelihood zones

Data

LSMS survey

- Ground truth data in Uganda/Tanzania/Malawi
- Three different rounds each with broad spatial coverage
- Household assets, demographics, geolocation of clusters
- Distance to roads and markets reflects access to market and information

Prices

- Major grain prices (maize, rice, nuts, beans etc.) in main agricultural markets prices, collectedly monthly
- Lagged one month before the survey time

Weather

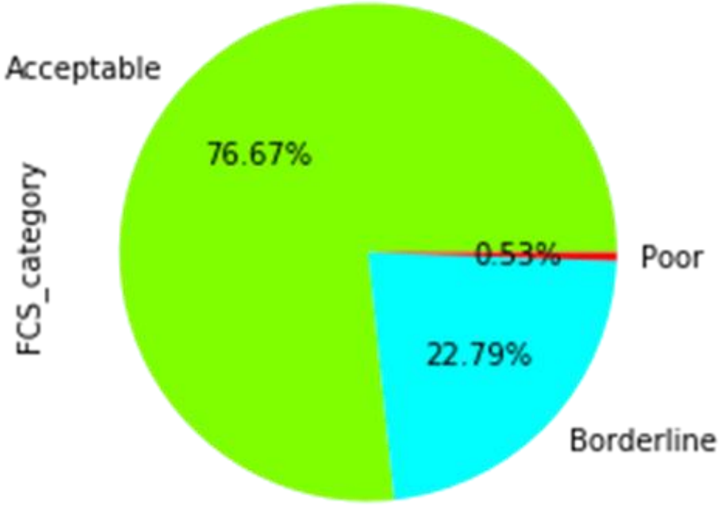
- Precipitation/temperature from remote sensing are relevant to agricultural production
- From the previous growing season

Categorical vs. Continuous

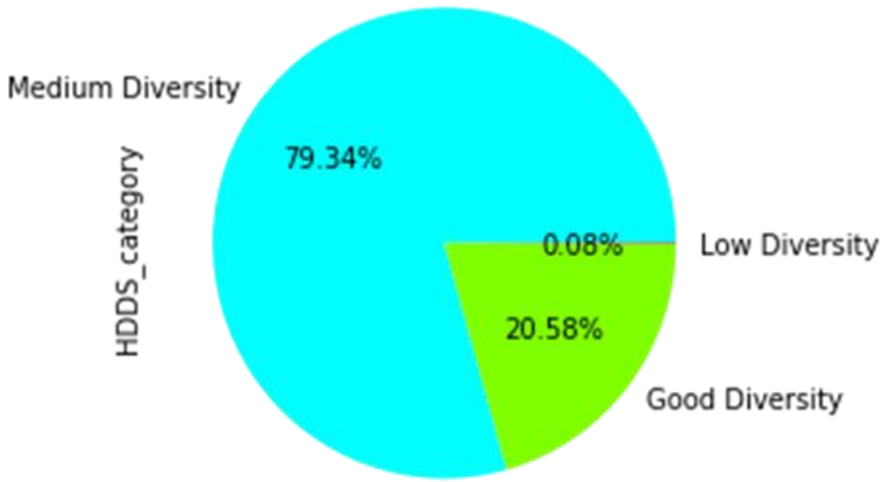
1. Focus on categorical prediction for the food security cutoffs
2. Policy-relevant
3. Classifiers are more sensitive to the majority class
4. Recall rate of the insecure villages is more important than accuracy
5. Apply down sampling, over sampling, and synthetic data techniques to force the model to learn about the tail of the distribution

Challenge

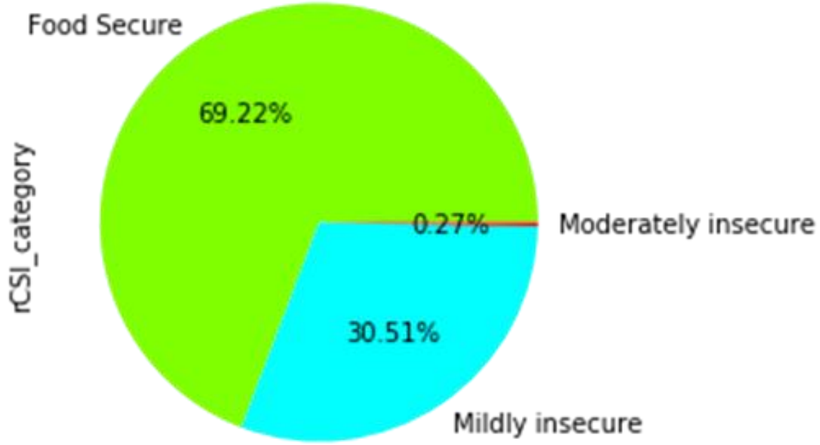
Detecting rare but relevant households



FCS



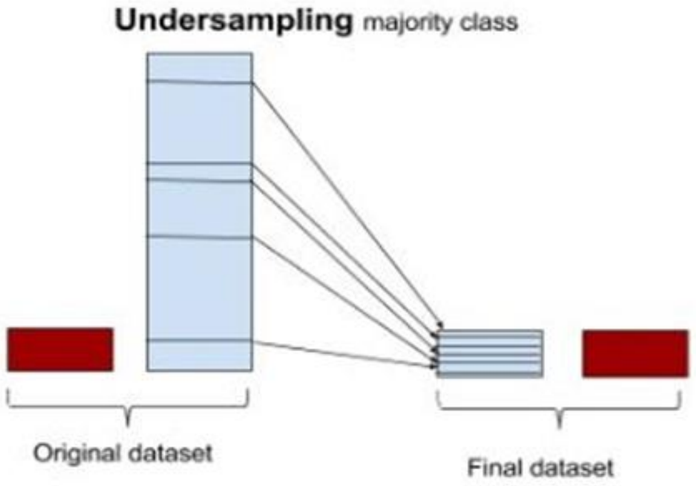
HDDS

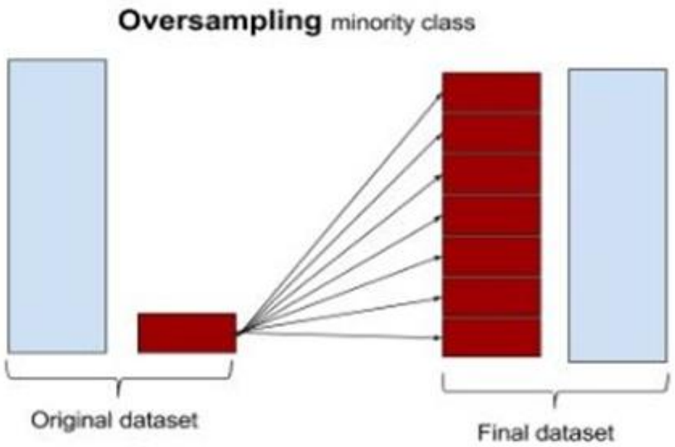


rCSI

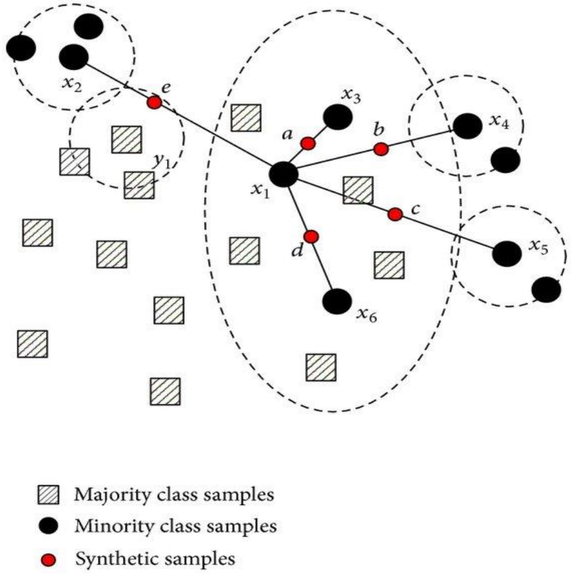
Method: sampling

Detecting rare but relevant households



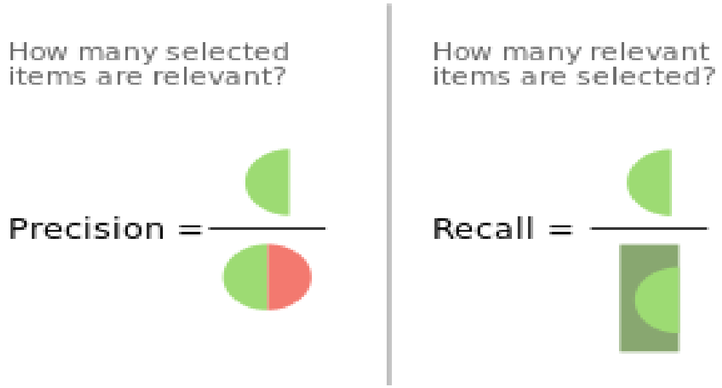


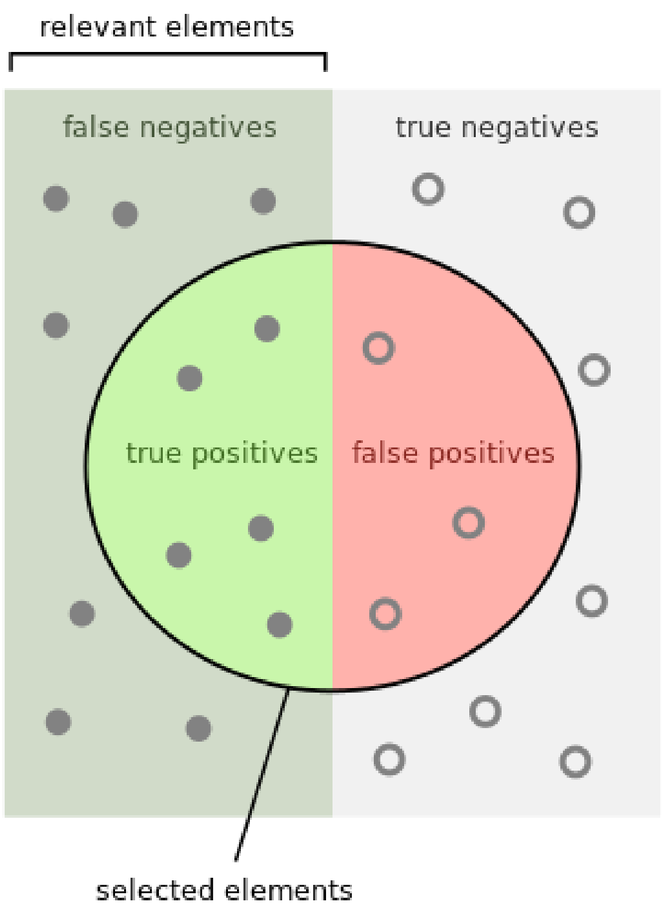
SMOTE and ADASYN



Results Metrics

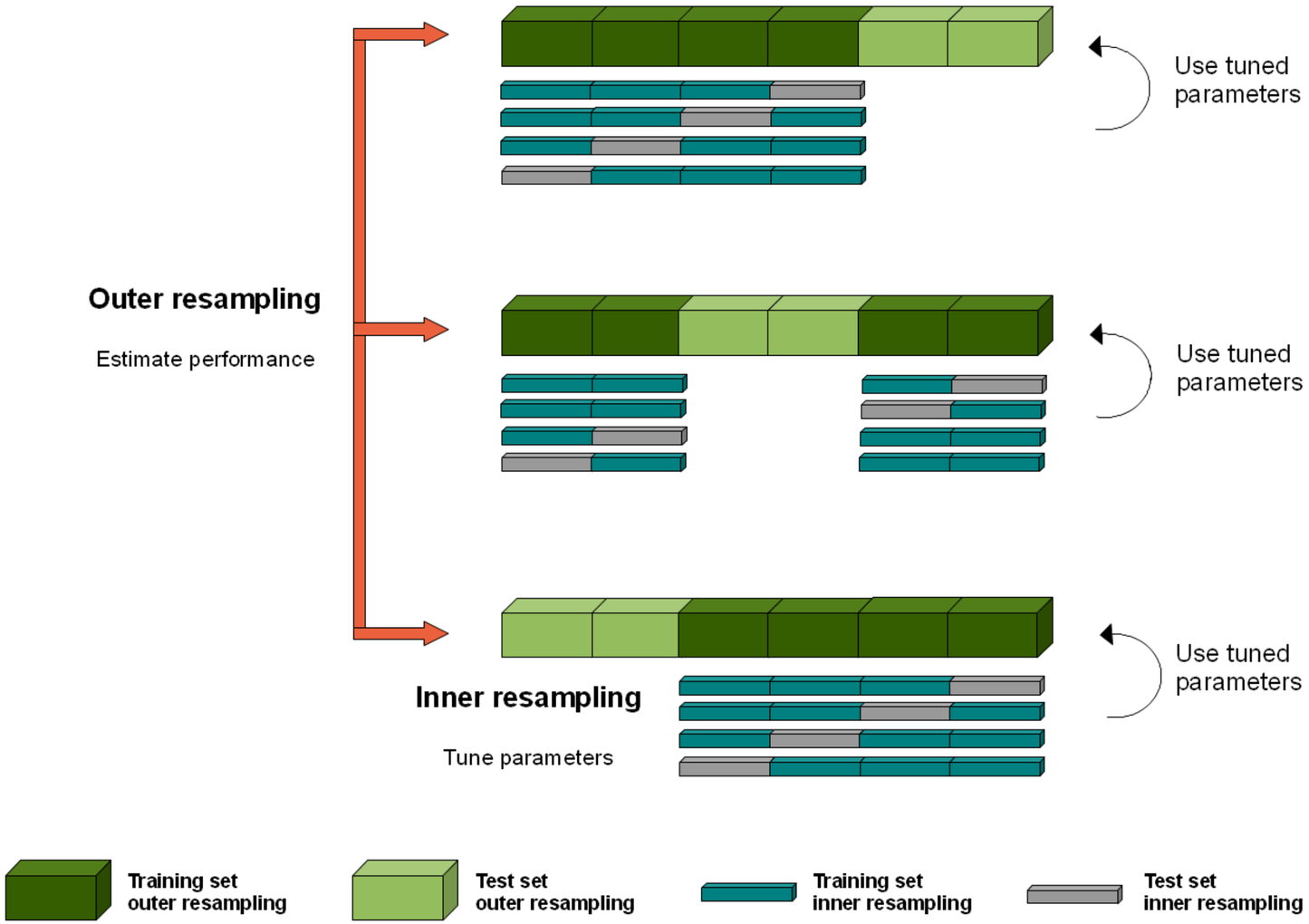
1. Recall (are we getting all the insecure households ?)
2. Precision (are we mistakenly categorizing secure households as insecure?)
3. f-1 score (balance recall and precision)
4. Overall categorical accuracy





Nested Cross-validation

- Rare events of food insecurity tend to vary a lot year by year, i.e. 1 or 2 cases in a good year vs > 50 cases in a bad year
- Use any two years as training data to predict the third year
- Inner-CV: tune parameters based on CV of each training set divided into K fold
- Outer-CV: Average out the performance after cross-validation to get more stable and trustworthy result



Classification Algorithm

0. **Logistic (baseline)**
1. **Classification Tree (base learner)**
2. **Random Forest (parallel)**
3. **Gradient boosting (sequential)**

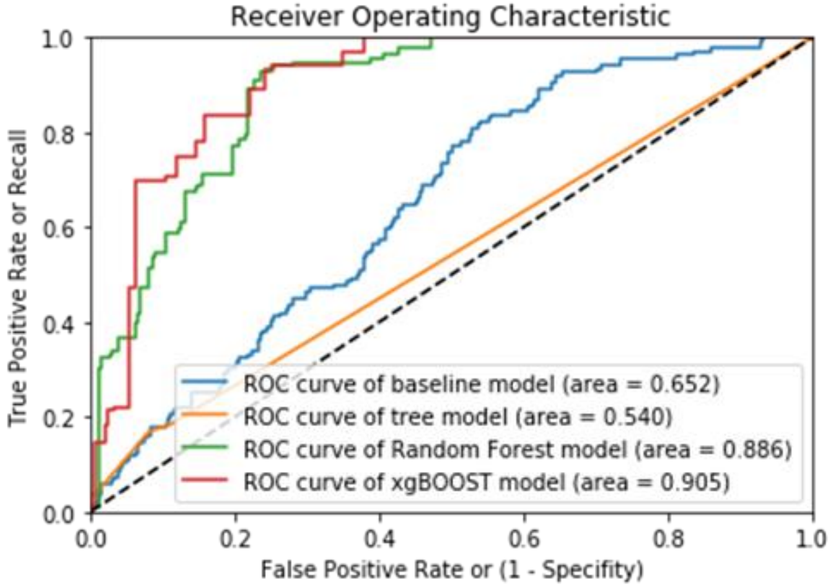
Compare to a Baseline

Logistic Regression
Data split: year split (cross-validated)
Data segmentation : by country
Down/over sampling: None

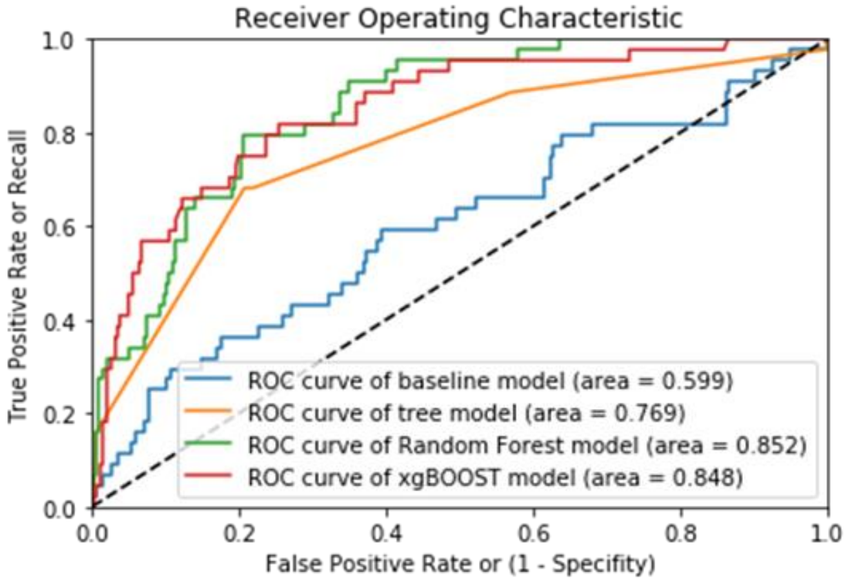
Variable groups:
Market: food price, market thinness
Asset: cellphone ownership, floor/roof material, asset index
Weather: dry spells, average temperature and rain
Location: elevation, distance to road, urban/rural
At village, district and regional level

Results for binary cut-off

FCS

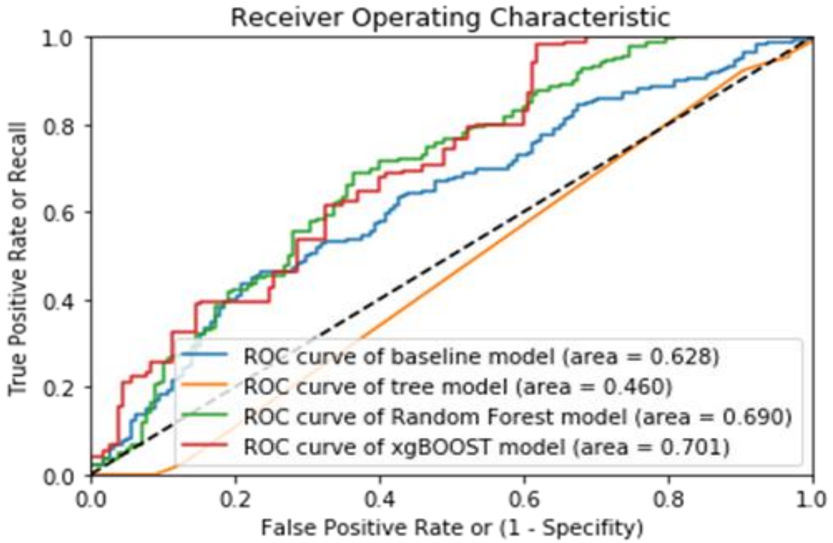


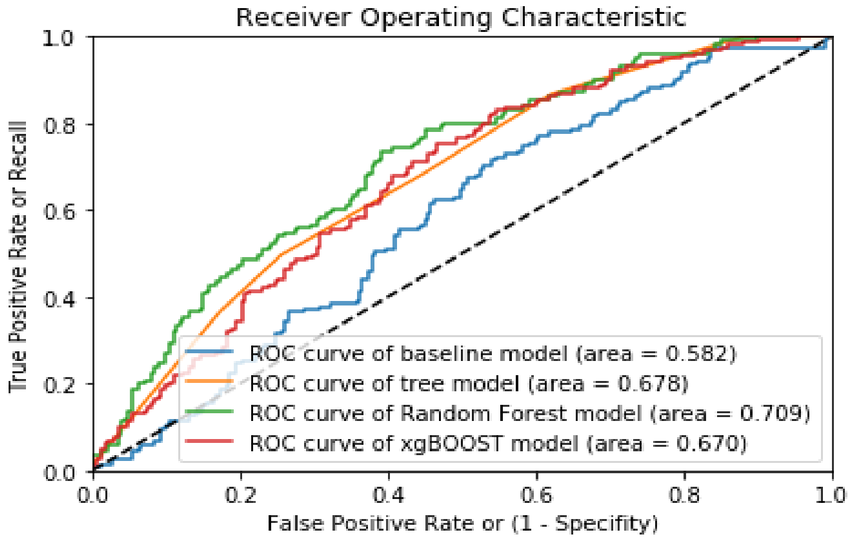
Malawi



Tanzania

rCSI





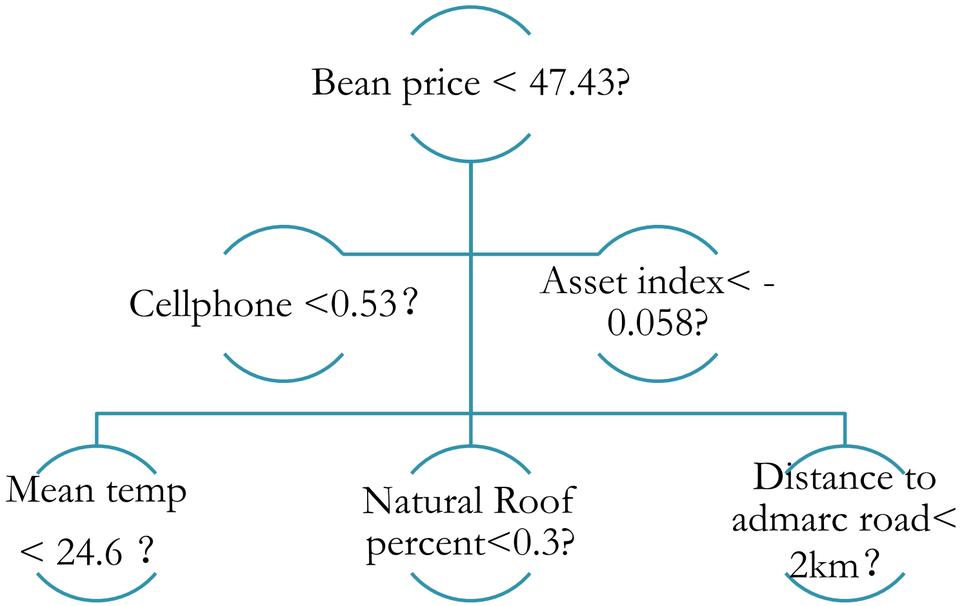
In table format…
Binary Baseline vs ML algorithms, no oversampling (year split)
Similar accuracy, higher recall

Country-Year (cross-validated)	Food Security Measure	Overall Accuracy (baseline)	Overall Accuracy (ML)	Recall Rate Insecure category (baseline)	Recall Rate Insecure category (ML)
Malawi	FCS	0.71	0.75	0.26	0.18
	rCSI	0.69	0.60	0.36	0.54
Tanzania	FCS	0.81	0.82	0.06	0.08
	rCSI	0.55	0.59	0.29	0.43
Uganda	FCS	0.67	0.59	0.36	0.33

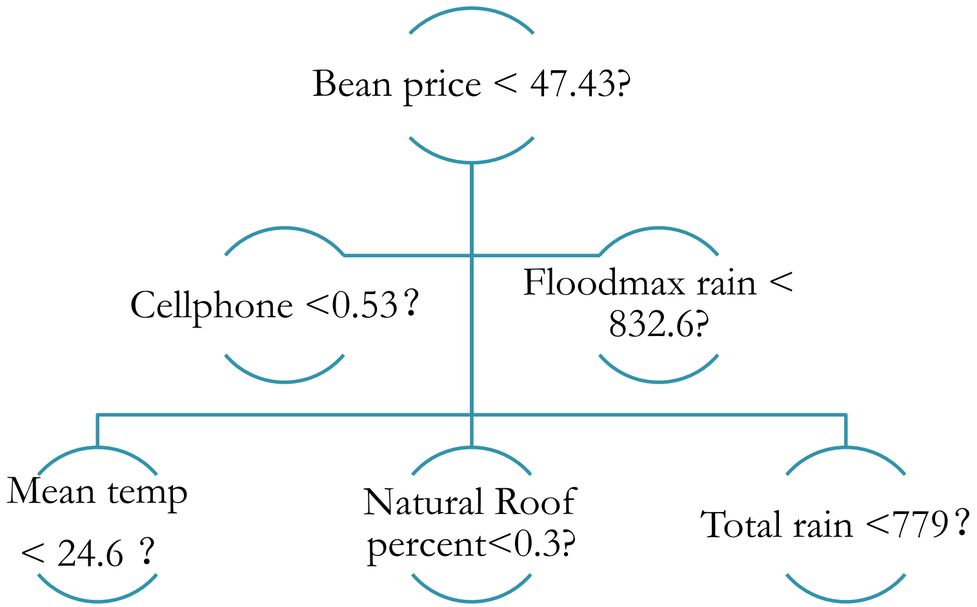
Table 2: Baseline vs ML algorithms with down/over sample technique

Country	Food Security Measure	Recall Rate Insecure category (Baseline)	Recall Rate Insecure category ML + Oversample	Recall Rate Insecure category ML + SMOTE-TOMEK	Recall Rate Insecure category ML + ADASYN
Malawi	FCS	0.26 / 0.00	0.32 / 0.27	0.26 / 0.06	0.38 /0.15
	rCSI	0.36 / 0.00	0.54 / 0.04	0.36 / 0.20	0.72 / 0.20
Tanzania	FCS	0.06 / 0.00	0.18 / 0.00	0.06 / 0.01	0.29 / 0.43
	rCSI	0.29 / 0.00	0.21 / 0.00	0.29 / 0.20	0.54 / 0.07
Uganda	FCS	0.37 / 0.00	0.31 / 0.00	0.36 / 0.37	0.36 / 0.30

Feature importance



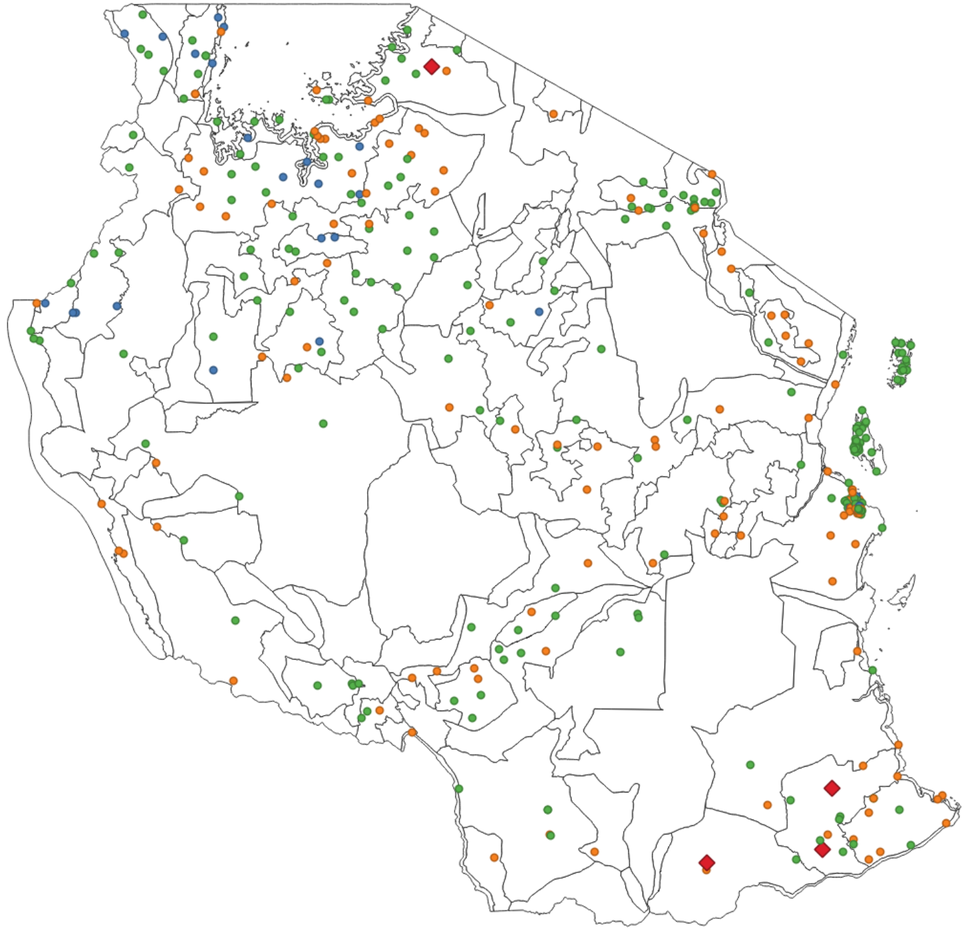
Original



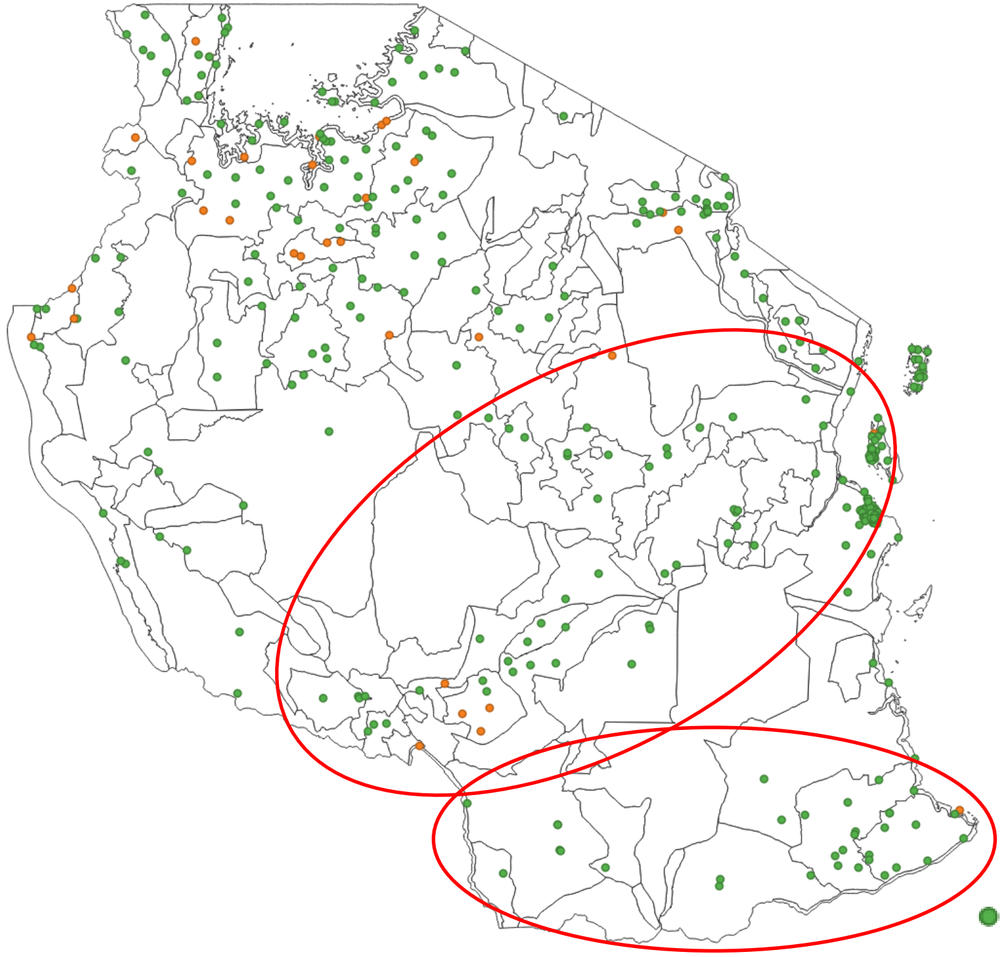
Oversample

Error Analysis

Tanzania rCSI in 2014/15



Baseline



ML+sampling

- Corret
- Overpredict by one
- Overpredict by two
- Underpredict by one
- Underpredict by two
- Livelihood zones

Ongoing work

- Price variable selection: which crop, where, and when
- Asset/roof variable prediction from satellite imagery
- Spatial-temporal correlation on data split
- Continuous approach: count of insecure HH for each cluster

Future Steps:

- Model generalization: models trained on one country/region to predict another
- Model interpretation: how much we gain from price and assets
- Prediction at "grid" level where satellite imagery available

Conclusions

1. Combined with data techniques, machine learning methods not only improve prediction accuracy in general, but particularly on households that are vulnerable to food price shocks.

2. An automated, updated and scalable food security system based on publicly available data, advanced data techniques can assist the work of food aid and humanitarian responses in a timely, transparent, and efficient fashion.

Chapter 3. Performance evaluation for forecasting modeling with spatiotemporal structures in data

Research Question

- What are the preferred cross-validation (CV) and out-of-sample (OOS) performance evaluation methods for given spatiotemporal correlations in panel data?
- How do we adjust for spatiotemporal correlations to improve the out-of-sample performance of the models?
- Furthermore, how can we tackle spatial and temporal heterogeneity?

Contributions

- First, we explore different spatial-temporal error structures in the panel dataset, how they affect the model performance, and model selection differently.
- Second, we showcase and measure the bias and apply the adjustment for spatial-temporal correlation in both environmental and development settings in real-world data.
- Third, we contribute to the econometrics literature on spatial panel forecasting with cross-validation strategies for evaluating model performance in a more robust way.
- Last, we provide modelling strategies for tackling spatiotemporal heterogeneity in the data.

Motivation - Problem

Great potential for ML in spatial-temporal data

Consequence of ignoring these correlations include the following (Robert et al., 2017):

1. Unreliable results: better than they are, making predictions less reliable.

2. Overfitting: prefer complex model

3. Misinterpretation of variables: The correlations can be incorporated into the model process through some of the covariates that also have spatial-temporal variation and appear to be more predictive than they actually are, e.g., rainfall.

Motivation - Solution

- Brenning (2012) uses k-means clustering to partition to reduce the influence of spatial autocorrelation.
- Roberts et al. (2017) invent "block cross-validation" by introducing spatial blocks on contiguous geographic space to force the model to be tested on more distant data, with similar ideas for temporal blocks of grouping data that fall in the nearby time interval.
- Meyer et al. (2018) create "Leave-Location-and-Time-Out (LLTO) CV" to address spatiotemporal correlations.
- Schratz et al. (2019) focus on the hyperparameter tuning aspect concerning the underlying spatial correlations in the data.

Methodology - Monte Carlo

Consider a Data Generating Process (DGP) in a panel setting with spatial-temporal correlations:

$$y_{it} = \gamma y_{it-1} + \rho \sum_{j=1}^{N} W_{ij} y_{it} + X_{it}\beta + \varepsilon_{it} \qquad (1)$$

where y_{it} is the dependent variable for region i at time t , X_{it} is a vector of explanatory variables, W_{ij} are elements of the spatial weight matrix W_N and ρ is the spatial lag coefficient, and ε_{it} is the error component with different specifications of spatially autocorrelated residuals.

Methodology - Monte Carlo

1. Define different spatial-temporal error structures, e.g. allowing for spatial diffusion patterns, in the case of air pollution

$$\varepsilon_{it} = u_i + \gamma_{it} + v_{it} \qquad (2)$$

u_i is an individual-specific time-invariant effect, γ_{it} follow a spatial diffusion process (correlated with γ_{it-1}) and a true random error v_{it}.

2. Demonstration of consequences for ignoring the spatial-temporal correlation
3. Adjustment for spatial-temporal error structures (k-means clustering, blocking)
4. Clustering methods for dealing with heterogeneity

Methodology - Empirical

Evaluate on two types of widely used panel datasets in development and environmental studies: household surveys and deforestation.

1. Assess spatial-temporal correlations in the data

2. CV and OOS performance comparison (on a correlated set and a relatively independent set)

3. Model implications (model complexity, variable selection, the spatial distribution of prediction error)

4. Adjustment for possible spatial-temporal correlations (k-means clustering, blocking)

5. Clustering methods for dealing with heterogeneity

Implications

- Show importance of adjusting for spatial and temporal correlation in development and environment application
- Propose more reliable result measurement for measuring and choosing the right model
- Implications on dealing with spatial and temporal heterogeneity

Q & A

Questions and comments?

Thank you!

zhou100@Illinois.edu